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Module 1: Supervised Learning

In supervised learning the training data is labelled: each example has an input and the correct output. Regression predicts a number and classification predicts a category. Linear regression fits a line by least squares, which minimises the sum of squared errors. Logistic regression passes a linear score through the sigmoid function to give a class probability. Decision trees split the data using information gain or Gini impurity. k-nearest neighbours classifies a point by the majority label among its k closest training points. A support vector machine finds the maximum-margin separating boundary. Overfitting happens when a model memorises the training data and performs poorly on new data. Regularisation fights it: L1 regularisation (lasso) penalises the absolute size of weights and can set some to zero, while L2 regularisation (ridge) penalises the squared size of weights.

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Module 2: Unsupervised Learning

Unsupervised learning finds structure in unlabelled data. The k-means algorithm picks k initial centroids, assigns each point to its nearest centroid, recomputes each centroid as the mean of its points, and repeats until assignments stop changing. The number of clusters k is often chosen with the elbow method, which plots the within-cluster sum of squares against k and looks for the bend. Hierarchical agglomerative clustering repeatedly merges the two closest clusters and is drawn as a dendrogram. DBSCAN groups points that are density-connected using two parameters, eps and minPts, and marks isolated points as noise. Principal component analysis reduces dimensions by projecting data onto the eigenvectors of the covariance matrix with the largest eigenvalues. The Apriori algorithm mines association rules using the measures support and confidence.

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Module 3: Model Evaluation

Data is split into training, validation and test sets; the test set is used only once at the end. In k-fold cross-validation the data is divided into k parts and each part is used once for validation; k = 10 is common. A confusion matrix counts true positives (TP), false positives (FP), true negatives (TN) and false negatives (FN). Precision = TP / (TP + FP). Recall = TP / (TP + FN). The F1 score is the harmonic mean of precision and recall. The ROC curve plots the true positive rate against the false positive rate, and the area under it (AUC) summarises the classifier. The bias-variance trade-off says that very simple models underfit (high bias) and very flexible models overfit (high variance). For regression, the mean squared error (MSE), its square root (RMSE) and R-squared are used. Data leakage, where test information reaches training, makes results look better than they are.